AI-POWERED EMERGENCY EVACUATION ROUTE PLANNING SYSTEM

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

During emergencies such as floods, fires, earthquakes, and industrial accidents, reaching a safe location quickly is critical. Normal navigation systems are mainly designed for everyday travel and may suggest routes that become unsuitable when roads are blocked or affected by an emergency. This creates a need for an evacuation-oriented route planning system that can consider the changing accessibility of roads and provide an appropriate alternative route.

The proposed **AI-Powered Emergency Evacuation Route Planning System** is designed to assist users in selecting an efficient evacuation route from their current location to a safe destination. The system represents the available road network as a weighted graph, where locations are treated as nodes and roads connecting them are treated as edges with associated travel costs. The user can select the required locations through the map interface and specify roads that are blocked or unsafe due to the emergency. These unavailable roads are excluded from the route calculation.

To determine the evacuation path, the system applies the **A* (A-Star) Search Algorithm**, an Artificial Intelligence search technique that combines the actual travel cost with an estimated distance to the destination. This enables the system to search the road network efficiently and identify a suitable available route. When a selected road becomes blocked, the system updates the road network and recalculates the route to provide an alternative path instead of directing the user through the affected area.

The main objective of the proposed system is to provide **dynamic and efficient evacuation route planning** based on the current availability of roads. By avoiding blocked routes and recalculating alternative paths, the system can reduce unnecessary delays and help users reach designated safe locations through available routes. The proposed approach also demonstrates how AI search techniques can be applied to real-world disaster management problems.

The system provides a foundation for intelligent emergency navigation and can be further extended with real-time emergency information, hazard-level identification, nearest emergency shelters, hospitals, and live road-condition updates.

Keywords: Artificial Intelligence, A* Search Algorithm, Emergency Evacuation, Route Planning, Dynamic Pathfinding, Graph Search, Blocked Roads, Disaster Management, Safe Route.

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Emergency situations such as floods and road blockages can make normal road routes unsafe. Natural disasters such as floods, fires, earthquakes and industrial accidents often require people to evacuate quickly, and during such events many roads may become blocked or unsafe, making it difficult to identify a secure evacuation route.

This project, titled “AI-Powered Emergency Evacuation Route Planning System”, uses OpenStreetMap road network data for Chennai [7]. The road network is represented as a graph of interconnected locations in which each road is assigned a travel cost. The A* (A-Star) Search Algorithm [13] is used to find safe evacuation routes, hazardous roads are avoided, and alternate routes are identified. The safe route is displayed on an interactive web map.

The project domain comprises Artificial Intelligence, Graph-Based Route Optimization and Geospatial Mapping. The application areas are Emergency Evacuation Planning, Disaster Management, Safe Route Navigation and Smart City Transportation.

The system helps to identify safer evacuation routes during emergencies and avoids blocked and hazardous roads while planning routes. It uses the A* Search Algorithm for efficient route finding, provides real-time visual route information through an interactive map, and supports faster and safer decision-making during emergency situations. The project was selected to develop a practical solution for safe evacuation route planning, to explore the use of AI and graph-based search for route optimization, and to apply real-world road network and hazard data to a real-world problem.

1.2 PROBLEM STATEMENT

Conventional navigation may not consider emergency-specific road hazards, and blocked roads can lead to unsafe or inefficient routes. There is therefore a need for a system that combines hazard information with route planning.

The problem needs to be addressed because unsafe routes during emergencies can delay evacuation, whereas avoiding hazardous roads can improve route safety and efficiency. Quick route selection is important for emergency response and evacuation. A system is needed to identify hazardous roads and avoid them. The proposed system uses the A* Search Algorithm to find alternate routes, and an interactive map provides a clear visual representation of the safe route.

1.3 OBJECTIVES

The objectives of the project are:

1. To develop an AI-based emergency evacuation route planning system for Chennai that helps users identify suitable evacuation routes using the Chennai road network.
2. To implement the A* Search Algorithm for finding an efficient and safe route between the selected source and destination.
3. To identify and avoid hazardous roads, preventing blocked or flooded roads from being used in the safe route.
4. To provide an alternate safe route when the original path contains a blocked or hazardous road.
5. To recalculate the route whenever road conditions change.
6. To reduce evacuation time by selecting an optimal path.
7. To visualize evacuation routes and hazards on an interactive web map.
8. To improve public safety during natural disasters.

1.4 SCOPE

The scope of the project is the planning of emergency evacuation routes on the Chennai road network obtained from OpenStreetMap. Users select a start location and a destination, either from the six predefined Chennai locations or by clicking a location marker on the map. Hazard information, namely blocked and flooded roads, is read from a CSV file (hazards.csv), and hazardous roads are removed from the routing network. The A* Search Algorithm then calculates a safe route, and the route and the hazards are displayed on an interactive web map. The system is applicable to Emergency Evacuation Planning, Disaster Management, Safe Route Navigation and Smart City Transportation. Comparison with existing methods, performance evaluation and the preparation of the final system documentation are planned as future work (Chapter 5).

1.5.SECURITY REQUIREMENTS/ SECURITY CONSIDERATION

- **Data Integrity:** Ensure that the road network and hazard data stored in hazards.csv are accurate and not modified by unauthorized users.
- **Input Validation:** Validate source and destination selections before calculating evacuation routes.
- **Hazard Data Validation:** Verify hazard entries to prevent invalid road-node pairs from affecting route planning.
- **Web Application Security:** Validate API requests and handle errors safely in the Flask web server.
- **Availability:** Ensure the application can access the stored road network and display routes when required.

1.6 SDG ALIGNMENT

The project contributes to the following Sustainable Development Goals (SDGs):

S. No.	SDG No. & Goal	Relevant SDG Target	Project Contribution / Alignment
1	SDG 11 – Sustainable Cities and Communities	Target 11.5 – Reduce the number of deaths and the number of people affected by disasters, including water-related disasters.	Provides safe evacuation routes in Chennai that avoid blocked and flooded roads, supporting safer and faster evacuation during emergencies and smart city transportation.
2	SDG 13 – Climate Action	Target 13.1 – Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters.	Supports disaster management by identifying hazardous roads, regenerating an alternate safe route when road conditions change, and visualizing the route on an interactive map.

Table 1.1 SDG ALIGNMENT

CHAPTER 2

LITERATURE REVIEW

2.1 EXISTING SYSTEM WITH SECURITY APPROACHES

The existing approach uses normal map-based route planning, in which a user selects a destination and the system provides the shortest route to it. The technologies and algorithms used are GPS, digital maps and Dijkstra or other shortest-path algorithms. Such systems are easy to use, perform fast route calculation and are easily available.



Figure 2.1 Existing System

Gaitanis et al. [1] (2023) proposed route planning for emergency evacuation using graph traversal algorithms, in which a floor-plan graph is generated and Dijkstra's Algorithm is used to find the shortest evacuation path. The work mainly focuses on the shortest path and is less suitable for changing emergency conditions. Dijkstra's Algorithm searches many nodes before reaching the destination, does not use heuristic information, requires higher execution time for large road networks, and its performance decreases as the graph size increases.

Huang et al. [2] (2024) presented a real-time automatic fire emergency evacuation route selection model based on Multi-Criteria Decision Making, which selects suitable evacuation routes by considering multiple emergency factors. The limitation is that the decision-making is more complex and may require more input information.

"Help Me Evacuate! A Smart Adaptable Evacuation System" [3] (2023) uses intelligent route planning to provide adaptable routes that support safe emergency evacuation, but it may have limited dynamic route recalculation for newly blocked roads. Other related studies include a route guidance map for emergency evacuation [4], integrated evacuation route planning based on real-time data acquisition [5], and sensing, calculating and disseminating evacuating routes during an indoor fire [6].

The limitations of the existing map-based systems are that they cannot handle sudden road blocks effectively, do not dynamically avoid flooded or hazardous roads, and may provide a route that becomes unsafe during emergencies. Security approaches used in the existing systems:

- **Secure Data Handling:** Existing navigation systems may use data validation to maintain the reliability of road and location information.
- **Access Control:** Some systems restrict access to sensitive information and administrative functions.
- **Secure Communication:** Systems may use HTTPS to protect data exchanged between users and servers.
- **Data Integrity:** Validation mechanisms help identify incorrect or inconsistent road and hazard information.
- **Privacy Protection:** Location data should be handled carefully to protect user privacy.

Note: The specific security approaches used by existing systems were not specified in the project materials. The points above are general examples and should not be presented as verified features of a particular existing system.

2.2 RESEARCH GAPS

The following research gaps were identified from the literature survey and the existing system:

- Conventional navigation does not consider emergency-specific road hazards and cannot handle sudden road blocks effectively.
- Shortest-path approaches mainly focus on the shortest path and are less suitable for changing emergency conditions.
- Dijkstra-based graph traversal does not use heuristic information, searches many nodes, and takes more time for large road networks and for route recalculation.
- Multi-criteria decision-making approaches are more complex and may require more input information.
- Intelligent route planning approaches may have limited dynamic route recalculation for newly blocked roads.

2.3 PROPOSED SYSTEM

2.3.1 OVERVIEW OF THE PROPOSED SYSTEM

The proposed system uses the A* Search Algorithm to calculate the safest evacuation route on the Chennai road network obtained from OpenStreetMap. Blocked and flooded roads are avoided. The key features of the system are hazard detection, a safe alternate route, an interactive map and emergency route planning. The working of the system is as follows:

1. The user enters the source location.
2. The user selects the destination.
3. Blocked roads are marked.
4. The road network is converted into a graph.
5. The A* Search Algorithm calculates the safest route.
6. An alternative path is generated if a road becomes blocked.
7. The final safe route is displayed.

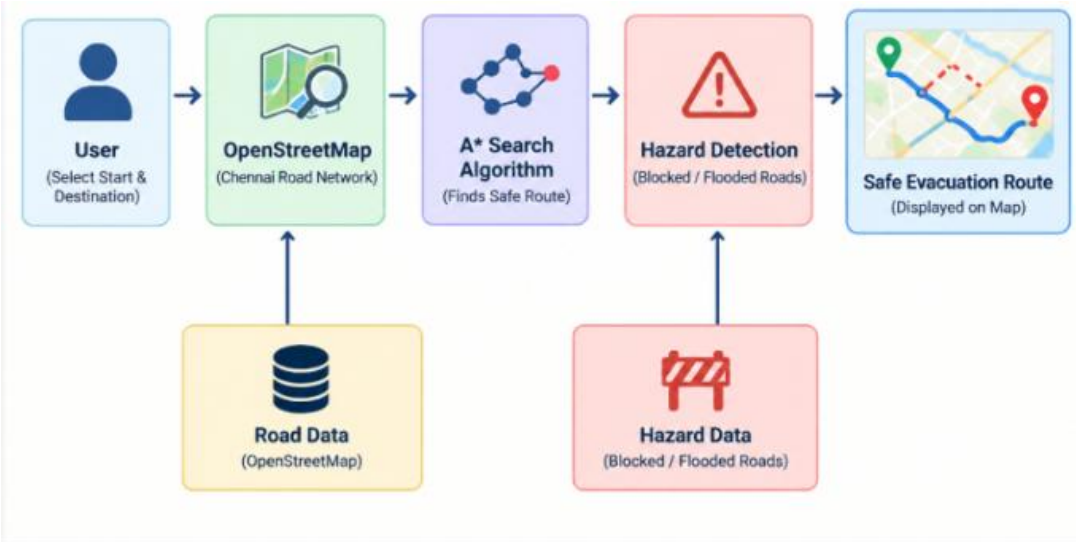


Figure 2.2 Proposed System

2.3.2 OBJECTIVES OF THE PROPOSED SYSTEM

The proposed system aims to avoid unsafe roads, to find an alternate route automatically when the normal route is affected, and to provide the route visually on the map. The expected outcomes are a safe evacuation route, faster route planning and reduced risk during emergencies.

2.3.3 ADVANTAGES OF PROPOSED SYSTEM

- **Real-time hazard-aware routing** – blocked and flooded roads are removed from the routing network before the A* search.
- **Dynamic rerouting** – when hazards change, a safe alternate route is generated automatically.
- **OSM-based real road network** – actual Chennai road network data is used rather than a fixed sample graph.
- **Visual hazard representation** – blocked roads, flooded roads and the safe A* route are shown directly on the map.
- **Integrated emergency decision system** – location selection, hazard detection, A* routing and the visual safe route are combined in one pipeline.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The system takes the selected locations and hazard information as input. The Chennai road network is obtained from OpenStreetMap and the locations are mapped to road nodes. The A* Search Algorithm then calculates a safe route while avoiding hazardous roads. Finally, the selected safe evacuation route is displayed on an interactive map. The architecture is shown in Figure 3.1.

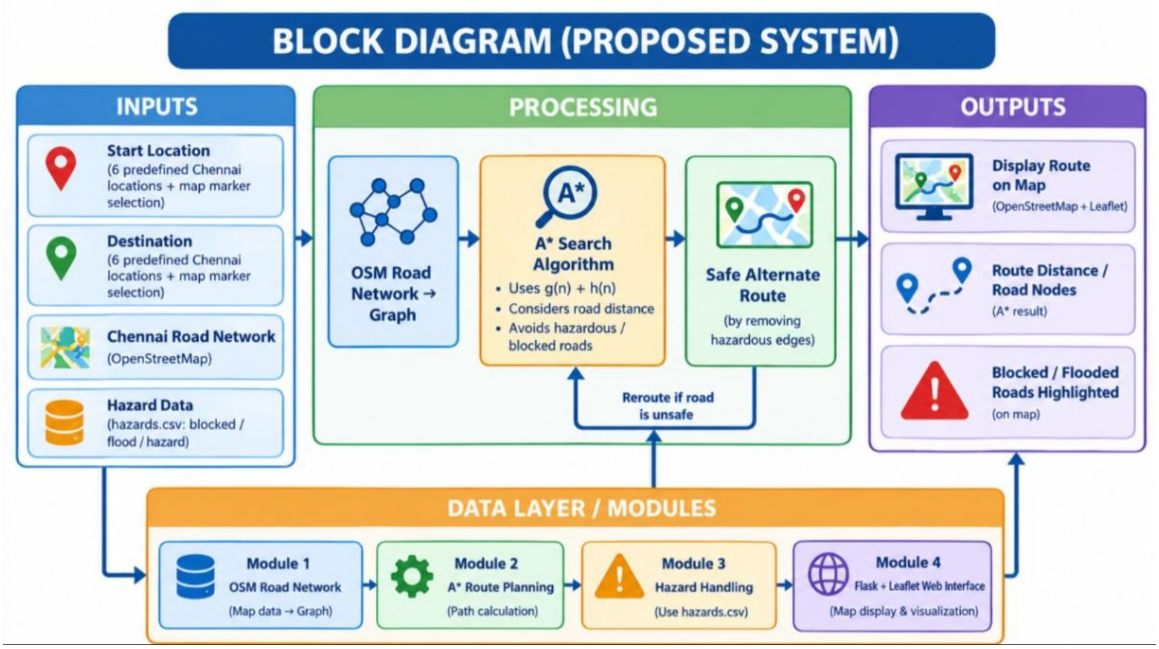


Figure 3.1 System Architecture

Inputs: the start location and destination (six predefined Chennai locations, or selection by map marker), the Chennai road network from OpenStreetMap, and the hazard data (hazards.csv, containing blocked and flooded roads).

Processing: the OSM road network is converted into a graph; the A* Search Algorithm uses g(n) + h(n), considers road distance and avoids hazardous or blocked roads; a safe alternate route is obtained by removing hazardous edges, and the route is recalculated if a road is unsafe.

Outputs: the route displayed on the map (OpenStreetMap and Leaflet), the route distance and road nodes (A* result), and the blocked and flooded roads highlighted on the map.

Hardware and software requirements: the hardware requirements are given in Table 3.1.

Component	Specification
Processor	Intel Core i3 or above
RAM	4 GB minimum
Storage	20 GB free space
Input devices	Keyboard and Mouse
System	Laptop/Desktop with Internet Connection

Table 3.1 Hardware Requirements

The software requirements are: Windows 10/11 operating system; Python programming language; Visual Studio Code / PyCharm as the IDE (Visual Studio Code and PowerShell as the tools used); the libraries Flask, OSMnx, NetworkX, Matplotlib and Leaflet; and Git as optional version control. The map data source is OpenStreetMap and hazard data is stored in CSV files.

3.2 MODULE DESIGN

The system is organised into five modules:

1. **Module 1 – OSM Road Network & Location Handling:** obtains the Chennai road network from OpenStreetMap and converts the map data into a graph; the selected locations are mapped to road nodes.
2. **Module 2 – A* Route Planning:** calculates the route between the selected source and destination using the A* Search Algorithm.
3. **Module 3 – Hazard Detection & Rerouting:** uses the hazard data (hazards.csv) to remove blocked and flooded roads from the routing network and finds an alternate route when the normal route is affected.
4. **Module 4 – Map Visualization & Web Server:** a Flask and Leaflet web interface that displays the map, the safe route and the hazardous roads.
5. **Module 5 – Emergency Route Output & Evaluation:** produces the safe evacuation route and evaluates it by route distance, number of nodes and safe route availability.

3.3 DATA FLOW/UML DIAGRAMS

The data flow of the system follows the stages Input, Preprocessing, Proposed Method/Model, Evaluation and Output, as shown in Figure 3.2. The data used and produced at each step is as follows:



Figure 3.2 Data Flow Diagram

1. Input: the Chennai road network, the selected locations and the hazard data.
2. Road network preparation and location-to-node mapping (OSM road network).
3. Road nodes, road distances and hazard information.
4. A* Search Algorithm with hazard avoidance (A* safe routing).
5. Evaluation: route distance, number of nodes and safe route availability (route and hazard check).
6. Output: the safe evacuation route displayed on an interactive web map.

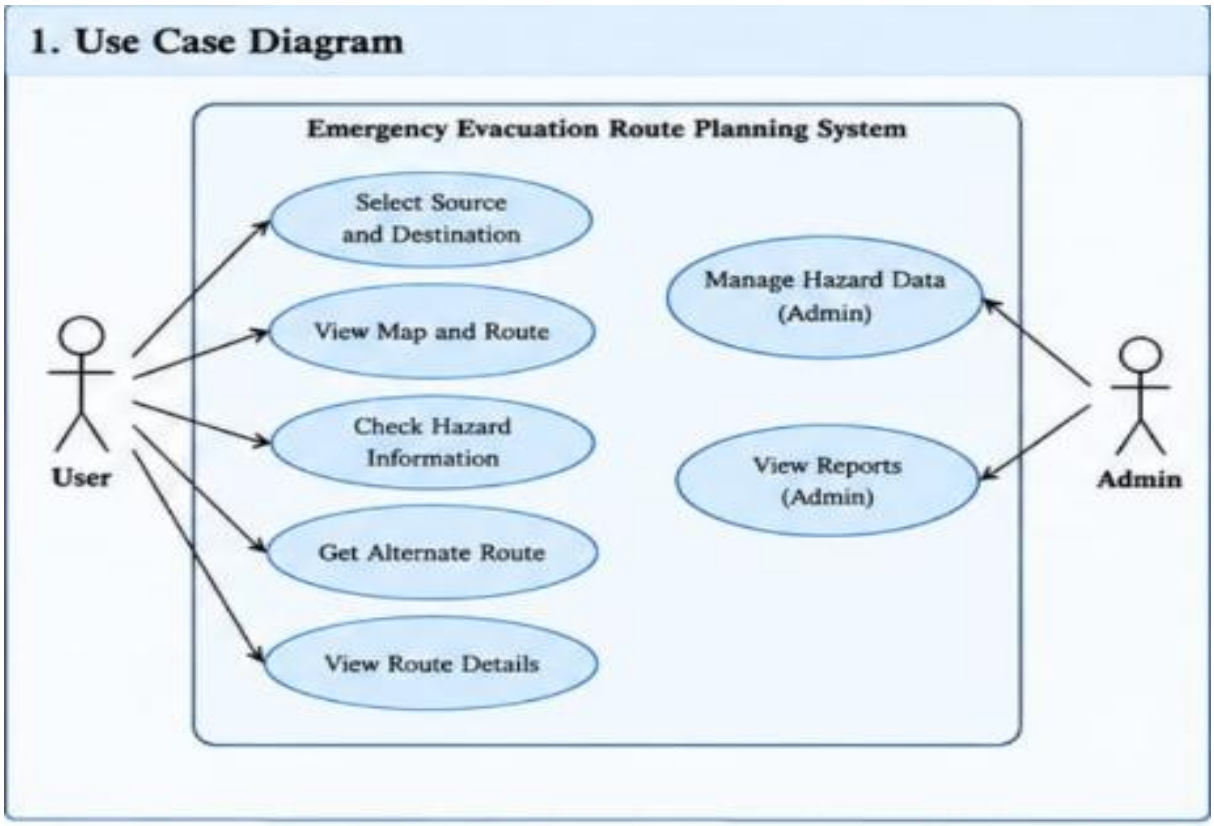


Fig 3.3 Use Case Diagram

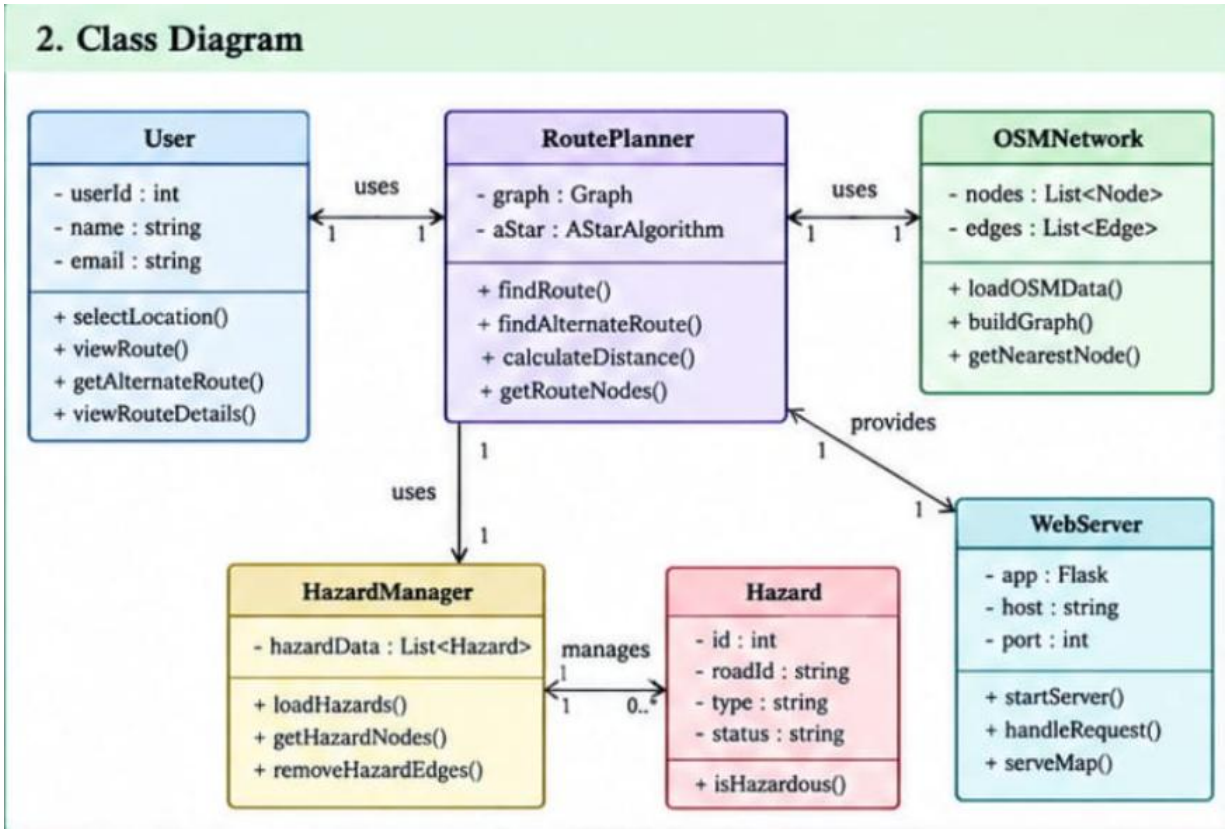


Fig 3.4 Class Diagram

3.4 THREAT MODEL AND ATTACK SURFACE

Threat Model

- False Hazard Data: Incorrect blocked-road or flood information may lead to unsuitable evacuation routes.
- Road Network Errors: Incomplete or outdated OpenStreetMap data may affect route accuracy.
- Unauthorized Data Modification: An attacker could modify hazards.csv and influence route planning.
- Route Manipulation: Altered source, destination, or routing requests could produce unexpected results.

Attack Surface

- Web Interface: Source and destination selection, route requests.
- Flask Server: API endpoints such as /api/route, /api/hazards, and /api/locations.
- Hazard Database: hazards.csv, containing road-node pairs and hazard status.
- External Map Data: OpenStreetMap road-network data.

Security Measures / Mitigation

- Validate source, destination, and hazard inputs.
- Restrict unauthorized modification of hazard data.
- Handle invalid requests and routing errors safely.
- Verify hazard information before using it for evacuation decisions.

3.5 SECURITY CONTROLS

- Input Validation: Validate source and destination selections before route calculation.
- Hazard Data Protection: Restrict unauthorized access and modifications to hazards.csv.
- API Validation: Validate requests received by the Flask server.
- Error Handling: Handle invalid inputs and route-not-found situations safely.
- Data Integrity: Verify hazard node IDs and statuses before using them for rerouting.
- Safe Route Verification: Check that the calculated route avoids the road edges marked

CHAPTER 4

SYSTEM DESCRIPTION

4.1 DATA COLLECTION

The data used by the system is collected from the following sources:

- **Road network data:** the Chennai road network is obtained from OpenStreetMap [7] using the OSMnx library [8].
- **Locations:** six predefined Chennai locations (for example, L6 – Guindy and L3 – Marina Beach) can be selected as the start location and the destination, or a location marker can be clicked on the map.
- **Hazard data:** blocked and flooded roads are maintained in a CSV file (hazards.csv). CSV files are used as the data store of the system.

4.2 DATA PREPROCESSING

Before route calculation, the road network is prepared and the selected locations are mapped to road nodes. The road network is converted into a graph, using NetworkX [9], consisting of road nodes and roads with their road distances as travel costs. The hazard information is then read from the hazard data, and the hazardous (blocked or flooded) roads are removed from the routing network so that they cannot be used in the safe route. The resulting graph is given to the A* Search Algorithm.

4.2.1 ALGORITHMS / METHODOLOGY

The AI technique used is the A* (A-Star) Search Algorithm [13], which finds an efficient route between the source and the destination by using the evaluation function $f(n) = g(n) + h(n)$, where $g(n)$ is the cost of the path from the start node to node n (road distance) and $h(n)$ is the heuristic estimate of the cost from node n to the destination. Because the heuristic guides the search, A* avoids the exhaustive exploration of many nodes that is characteristic of Dijkstra's Algorithm in the existing system.

The methodology followed is:

1. The Chennai road network, the selected locations and the hazard data are taken as input.
2. The road network is prepared and the locations are mapped to road nodes.
3. Road nodes, road distances and hazard information are used to build the routing graph.
4. The A* Search Algorithm with hazard avoidance calculates the safe route; if a road becomes blocked, an alternative path is generated.
5. The route distance, the number of nodes and the safe route availability are evaluated.

6. The safe evacuation route is displayed on an interactive web map.

The technologies used are Python [10] as the programming language; Flask [11], OSMnx [8], NetworkX [9] and Leaflet [12] as frameworks and libraries; CSV files as the database; OpenStreetMap as the map data; and VS Code and PowerShell as the tools.

4.2.2 RESULTS AND DISCUSSION

The web interface provides drop-down lists to select the Start Location and the Destination, the buttons “Find Safe Route” and “Clear Route”, and the option to click a location marker on the map to select the start and destination. Figure 4.1 shows the result for the start location L6 – Guindy and the destination L3 – Marina Beach, where the system displays the message “Safe evacuation route found successfully!” and draws the route on the OpenStreetMap-based map.

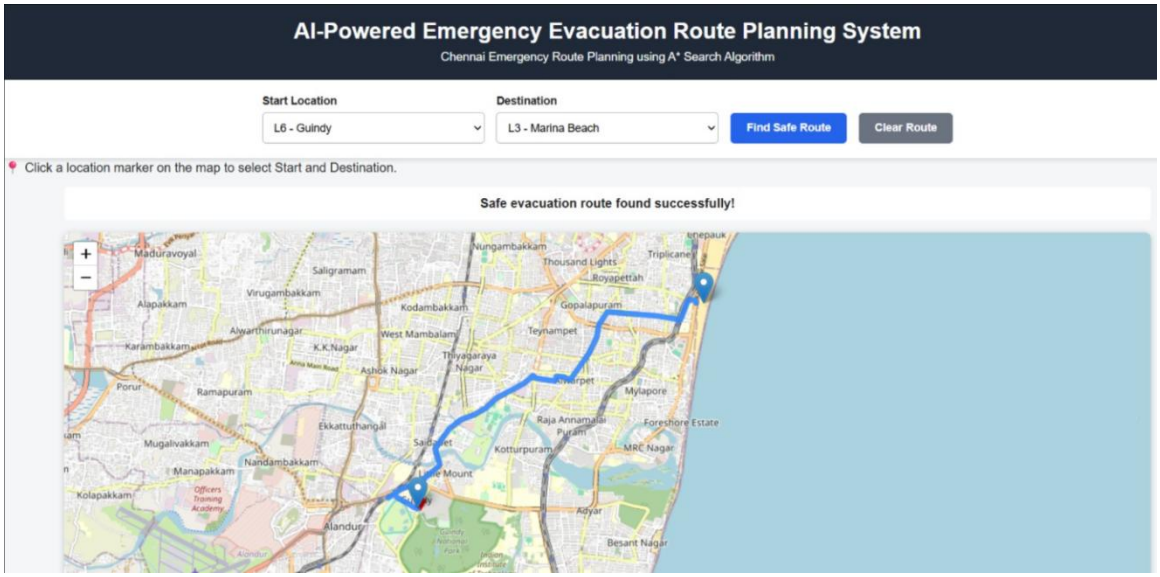


Figure 4.1 Safe Route Screenshot

Figure 4.2 shows a zoomed view of the map in which a blocked road is highlighted on the map along with the displayed route. In the system, blocked roads are shown in red, flooded roads in orange and the safe A* route in blue. The normal route and the hazard-affected route were tested successfully in the preliminary work.

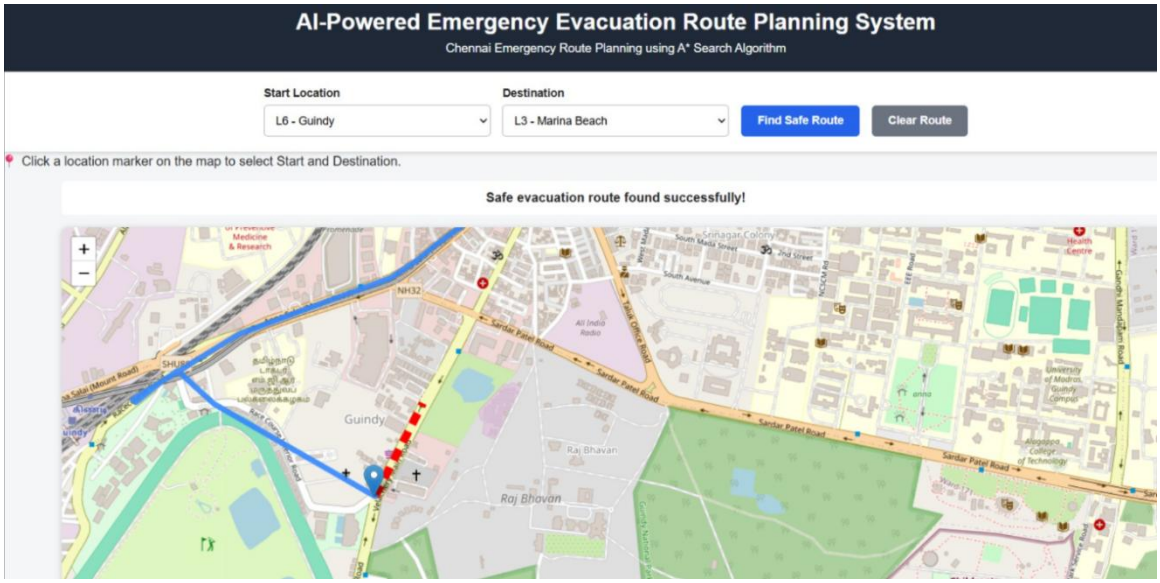


Figure 4.2 Blocked Road Highlighted on the Map

Metric	Normal A* Route	After Hazard
Start Location	Chennai Central	Chennai Central
Destination	Government General Hospital	Government General Hospital
Number of Nodes	23	25
Route Distance	2,320.95 m	2,423.04 m
Hazard Avoidance	No hazard removed	Blocked road avoided

Table 4.1 Route Planning Results

Existing Approach	Proposed System
Normal route selection	Hazard-aware route planning
May not account for manually recorded hazards	Uses hazard information from hazards.csv
Alternate routing depends on the navigation system	Re-runs A* after hazardous edges are removed

Table 4.2 Comparison with Existing Approach

CHAPTER 5

CONCLUSION

5.1 CONCLUSION

The proposed system successfully combines OpenStreetMap road data and the A* Search Algorithm to plan emergency evacuation routes. It maps the selected locations to the road network and calculates a suitable route between the source and the destination.

The system also considers hazard information such as blocked roads and removes hazardous roads from the routing network. This allows the system to generate an alternate safe route when the normal route is affected.

Finally, the generated route is displayed through an interactive web map. Thus, the system provides a simple approach for visualizing safe evacuation routes during emergency situations.

As future work, the remaining modules are to be completed, the system is to be compared with the existing routing approach, the route distance, nodes and safe rerouting performance are to be evaluated, and the final system screenshots, results and documentation are to be prepared.

5.2 FUTURE WORK

- **Real-Time Hazard Integration:** Integrate real-time flood, road blockage, and traffic information for improved route planning.
- **Multiple Hazard Handling:** Extend the system to avoid multiple blocked or hazardous road segments simultaneously.
- **Expanded Location Coverage:** Include more locations and road networks across Chennai.
- **Travel-Time Optimization:** Consider traffic conditions, road accessibility, and hazard severity along with distance.
- **Mobile Application Development:** Develop a mobile-friendly application for easier access during emergencies.
- **Performance Evaluation:** Evaluate route distance, computation time, and rerouting efficiency under different emergency scenarios.

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APPENDIX

A. Source Code Listings

- modules/osm_network.py – Chennai road network download and storage.
- modules/location_handler.py – Selected locations mapping to OSM nodes.
- module_2/astar.py – A* route planning algorithm.
- module_3/hazard_rerouting.py – Hazard-aware alternate route generation.
- module_4/map_server.py – Flask web server and API endpoints.
- module_4/web/index.html – Web interface.
- module_4/web/map.js – Interactive map, route and hazard visualization.

B. Project Structure

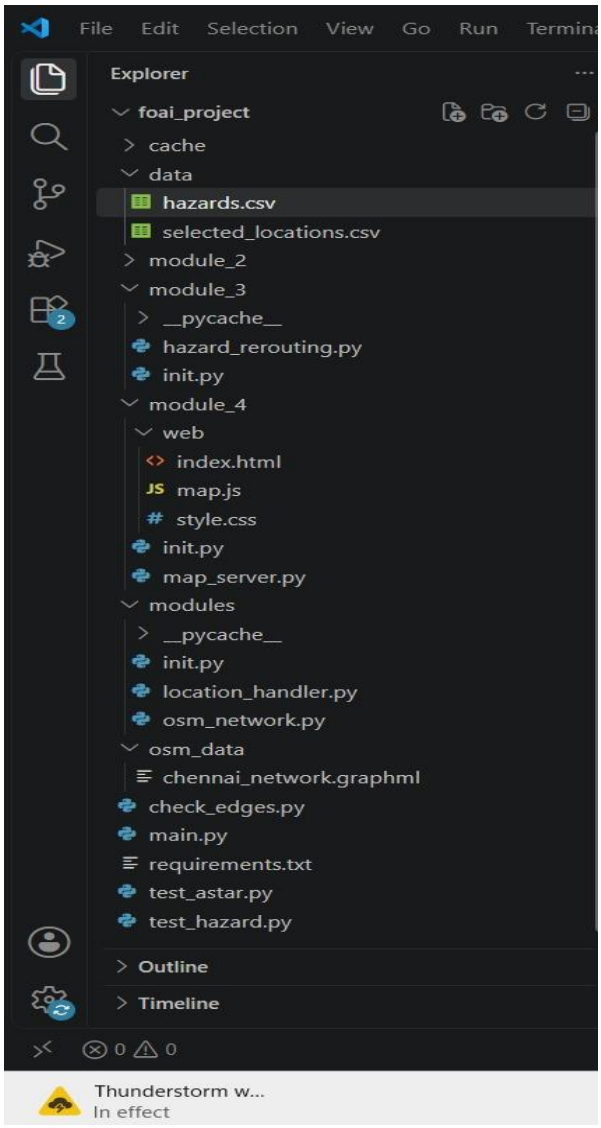


Fig B.1 Project Structure

C. Screenshots

C.1 Selected Locations.

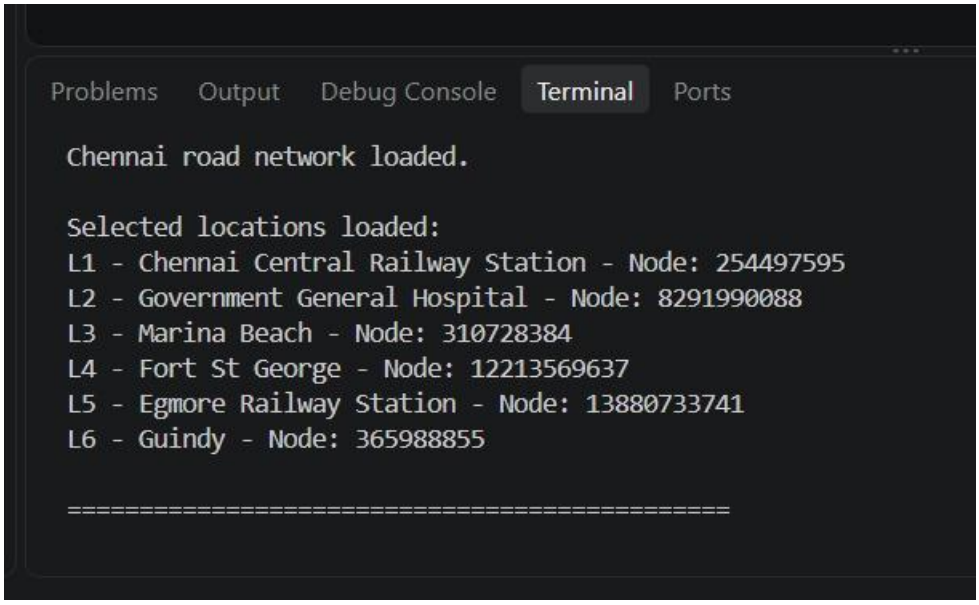


Fig C.1 Selected Locations

C.2 Locations and their mapped node IDs.

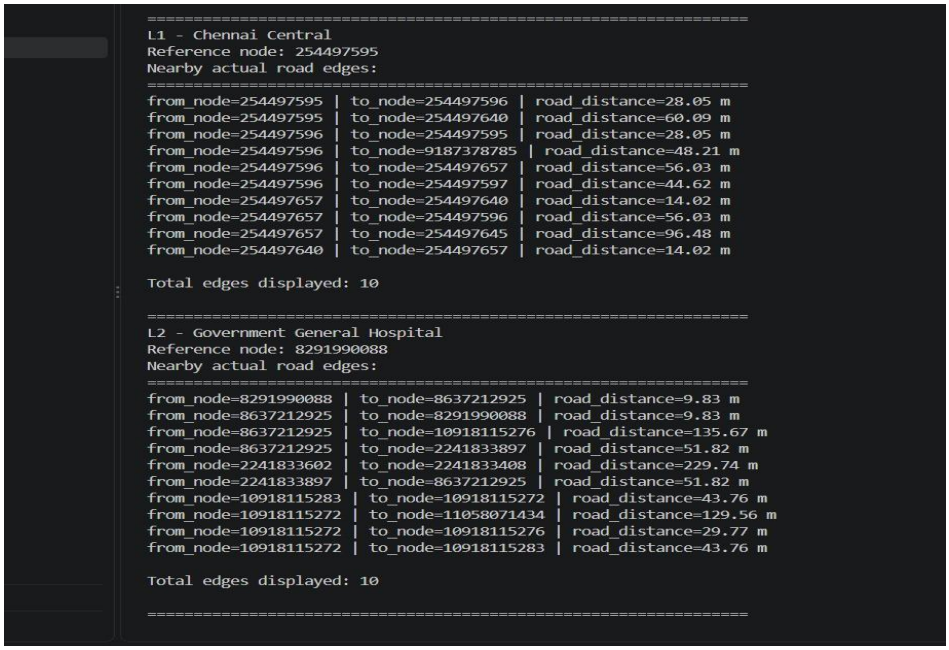


Fig C.2 Mapped nodes of L1 and L2

```
================================================================
L3 - Marina Beach
Reference node: 310728384
Nearby actual road edges:
================================================================
from_node=310728384 | to_node=243066060 | road_distance=12.75 m
from_node=310728384 | to_node=310728524 | road_distance=426.83 m
from_node=243066060 | to_node=259058458 | road_distance=83.80 m
from_node=243066060 | to_node=10277860575 | road_distance=55.25 m
from_node=310729326 | to_node=310728384 | road_distance=85.30 m
from_node=259058458 | to_node=310729326 | road_distance=8.97 m
from_node=259058458 | to_node=1279571365 | road_distance=165.35 m
from_node=243066062 | to_node=259058458 | road_distance=21.05 m
from_node=243066062 | to_node=2217244411 | road_distance=63.01 m
from_node=10277860575 | to_node=12123477279 | road_distance=16.14 m

Total edges displayed: 10

================================================================
L4 - Fort St George
Reference node: 12213569637
Nearby actual road edges:
================================================================
from_node=12213569637 | to_node=9659614459 | road_distance=874.33 m
from_node=12213569637 | to_node=12213569640 | road_distance=90.86 m
from_node=12213569649 | to_node=12213569648 | road_distance=29.26 m
from_node=12213569649 | to_node=12213569637 | road_distance=36.37 m
from_node=12213569648 | to_node=12213569649 | road_distance=29.26 m
from_node=12213569648 | to_node=12213569640 | road_distance=26.54 m
from_node=12213569640 | to_node=12213569648 | road_distance=26.54 m
from_node=12213569640 | to_node=12213569637 | road_distance=90.86 m
from_node=30237377 | to_node=258987179 | road_distance=121.78 m
from_node=30237377 | to_node=2271543502 | road_distance=586.64 m

Total edges displayed: 10
```

Fig C.3 Mapped nodes of L3 and L4

```
================================================================
L5 - Egmore Railway Station
Reference node: 13880733741
Nearby actual road edges:
================================================================
from_node=13880733741 | to_node=10945638487 | road_distance=7.07 m
from_node=10945638487 | to_node=13880733755 | road_distance=13.34 m
from_node=10945638487 | to_node=2198125458 | road_distance=186.77 m
from_node=13880733755 | to_node=13880733762 | road_distance=3.27 m
from_node=13880733762 | to_node=2197639805 | road_distance=13.92 m
from_node=13880733762 | to_node=2482057614 | road_distance=262.40 m
from_node=10945638478 | to_node=10945638504 | road_distance=27.33 m
from_node=10945638478 | to_node=13880733741 | road_distance=23.62 m
from_node=2197639805 | to_node=10945638478 | road_distance=11.53 m
from_node=10945638495 | to_node=13880733741 | road_distance=28.47 m

Total edges displayed: 10

================================================================
L6 - Guindy
Reference node: 365988855
Nearby actual road edges:
================================================================
from_node=365988855 | to_node=365988848 | road_distance=11.27 m
from_node=365988855 | to_node=7826160038 | road_distance=641.05 m
from_node=365988855 | to_node=6444634790 | road_distance=302.08 m
from_node=365988848 | to_node=365988855 | road_distance=11.27 m
from_node=365988848 | to_node=11668396343 | road_distance=232.27 m
from_node=11668396343 | to_node=11100965248 | road_distance=95.94 m
from_node=11668396343 | to_node=11668396344 | road_distance=70.27 m
from_node=11668396344 | to_node=11668396343 | road_distance=70.27 m
from_node=6444634789 | to_node=6444634790 | road_distance=14.53 m
from_node=6444634789 | to_node=365988848 | road_distance=297.50 m

Total edges displayed: 10
PS C:\Users\ramyadhivya\OneDrive\Documents\Desktop\foai_project>
```

Fig C.4 Mapped nodes of L5 and L6

D. Screenshot for Output

D.1 Blocked road highlighed in red.

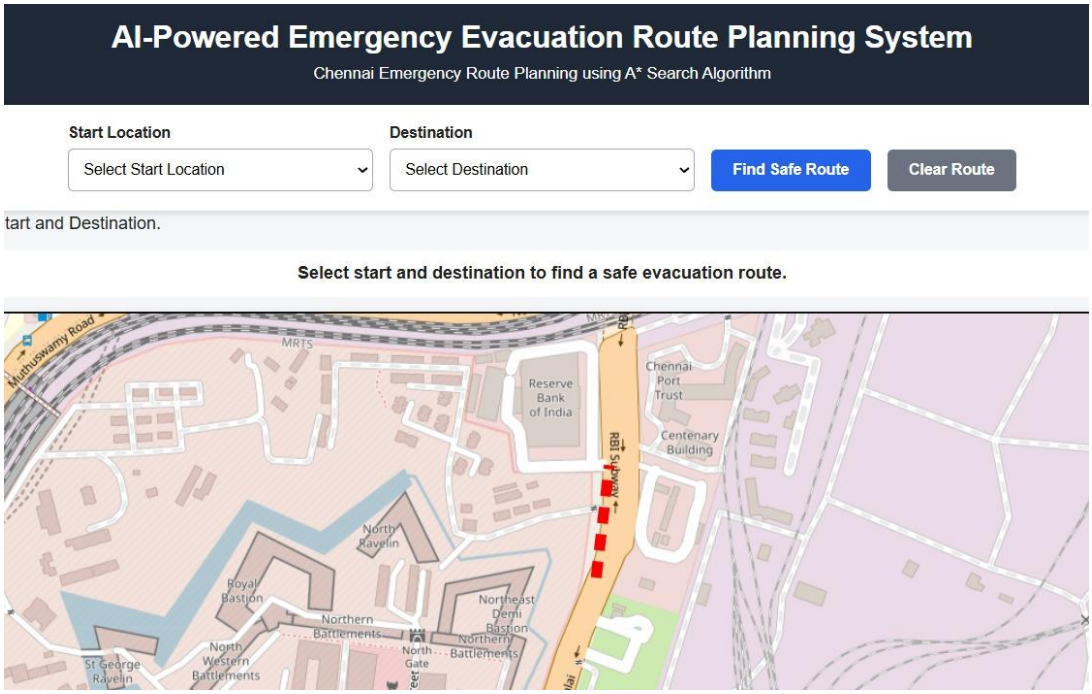


Fig D.1 Blocked road

D.2 Flooded road highlighted in orange.

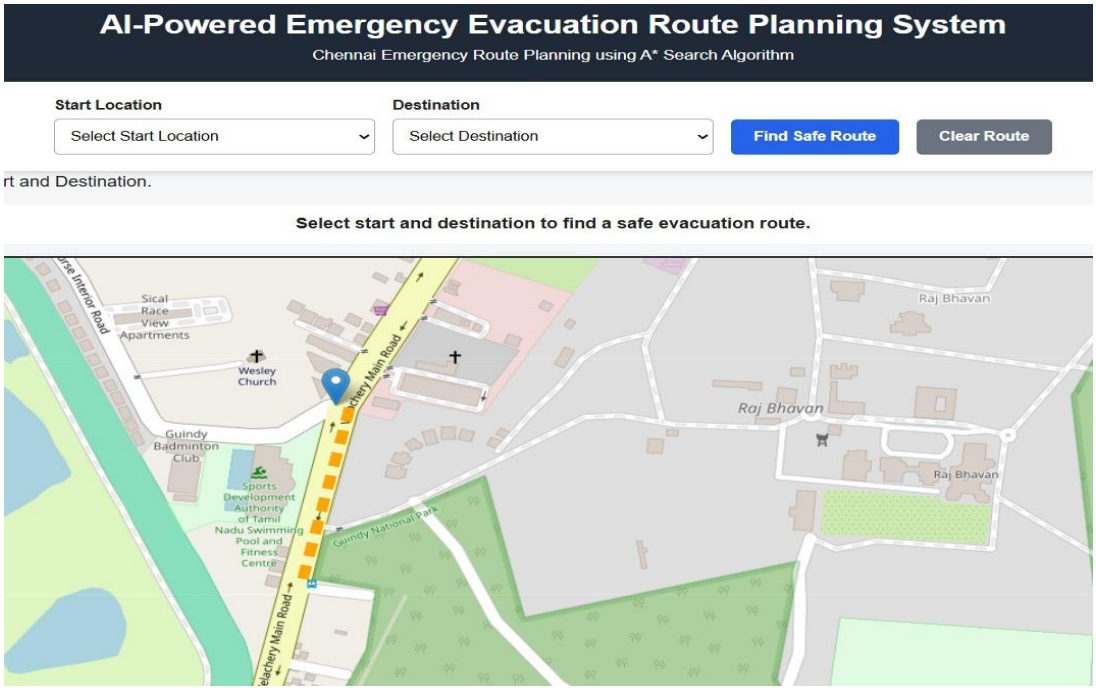


Fig D.2 Flooded road

D.3 Safe route displayed after a hazard is applied.

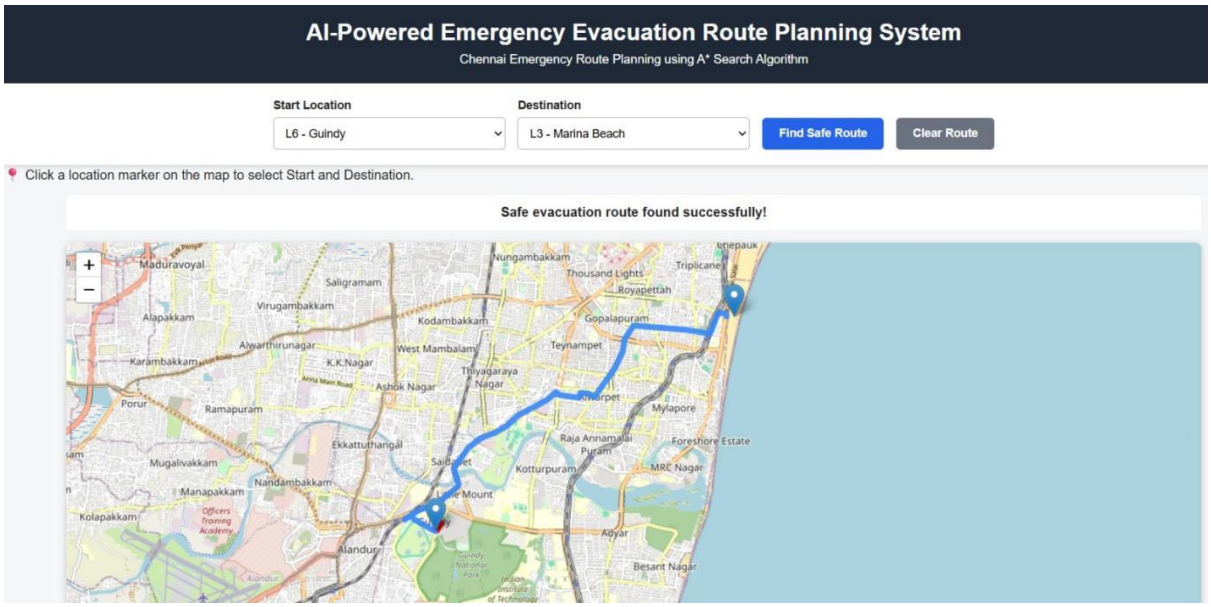


Fig D.3 Safe Route